AI-Powered Hospital Incident Management & Patient Safety Improvement

From incident report to approved corrective action - with a human in control of every release.

Business pitch | GAHAR Hospital Accreditation Standards 2025 alignment
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Proposal stage: targets are not results; all data synthetic.

Incident follow-up is slow, fragmented and hard to prove

Who has the problem. Quality and patient-safety officers in a general hospital receive incident reports from many departments. Forms arrive incomplete, are triaged by hand, and corrective actions sit on separate lists with no visible owner.

What it costs today

- **Time:** about **20 minutes** to prepare a first-pass summary per incident (planning assumption - to be timed locally before the checkpoint).
- **Risk:** missing details and inconsistent escalation delay review; actions without owners or due dates are not followed up.
- **Accreditation:** GAHAR expects serious events to be notified within 48 hours and an RCA with a corrective plan within 45 days - hard to meet with manual tracking.

Where it sits in the patient journey

Across admission, treatment, procedures and follow-up: any setting can raise an incident; the quality office turns them into reviews and tracked actions.

Current workaround: manual spreadsheet review, chat messages and separate action lists - slow, inconsistent, and hard to audit.

Three groups, one clear daily user

Role	Who	What they want
Operator	Quality and Patient Safety Officer	Fewer hours cross-checking forms; traceable, safe escalation; clear owners and deadlines.
Beneficiaries	Clinical teams, staff, patients, hospital management	Faster learning from incidents; fewer repeat events; calmer accreditation surveys.
Buyer	Hospital quality director / executive administration	Measurable time saved, auditable approvals, readiness against GAHAR expectations.

Persona

A quality officer handling reports from several departments, today chasing incomplete forms and owners by hand. She values **traceability, speed and safe escalation**, and will not accept a tool that acts without her approval.

One integrated workflow, five connected stages

Backend: authorized quality-report mail / signed webhook | Action register | Monitoring dashboard | Failed-run alerts

- **Data collection starts the run:** an incident form (or a new sheet row) triggers the workflow automatically.
- **Backend receives the authorized quality report by email:** only listed reviewers with the one-time code can approve; everything else is logged and alerted.
- **Grounded in GAHAR:** the 2025 hospital handbook is loaded as the knowledge base; unverified references are flagged, never invented.
- **Built in n8n** as a single workflow file, so the team can import, demo and version it in one step.

One synthetic incident, from submission to action

Step	What the audience sees	What it proves
1	Officer-friendly form: a medication near miss is submitted (no harm).	Data collection starts the workflow; no copying between tools.
2	Telegram / sheet shows: screened clean, routed to the clinical specialist.	PHI gate ran before any AI call; classification is explainable.
3	Draft RCA: confirmed facts, hypotheses, missing evidence, actions with role owners.	Evidence-limited: it does not invent injury or blame staff.
4	Judge scores and GAHAR citation status appear; unverified references are flagged.	Quality is measured on every draft.
5	Quality officer replies to the email: APPROVED + code + owner + due date.	Human approval is mandatory and authenticated.
6	Action register shows APPROVED / OPEN; dashboard counters update.	Follow-up is visible and measurable.
7	Contrast case: a spoofed reply from an unlisted address is rejected with an alert.	Fail-closed security; nothing is released.

Rehearse until it runs identically; keep the n8n canvas on screen only briefly; keep a recorded backup run ready.

Targets to prove - actuals reported honestly after testing

KPI	Lens	Baseline (provisional)	Target	How measured	Actual
Draft preparation time	Cost	20 min (assumed)	10 min or less	Time 10 synthetic reports before vs after	Pending
Routing accuracy	Outcomes	Not measured	90% or more	Gold-labeled benchmark; uncertain cases escalate	Pending
High-risk approval	Outcomes	Not measured	100%	Audit approval log on high-risk cases	Pending
Action owner coverage	Outcomes	Not measured	100%	Approved records with owner and due date	Pending
Officer effort	Well-being	Not measured	4 of 5 or higher	Short post-test survey	Pending

Business value (illustrative, if the target is met)

Saving 10 minutes on each of 100 reports is about **17 hours** of officer time - arithmetic from the assumed baseline and target, not a measured result. The larger value is control: every high-risk release is approved by a named person, and each action has an owner and a date.

AI drafts, a qualified human decides

Control	How it works
Synthetic data only	No live patient records, names, phone numbers or national IDs in prompts, logs, slides or demos.
PHI and injection gate	Regex scan before any model call; flagged reports are quarantined and logged in redacted form; output is scanned again before notification.
Human approval	No actions are released without a reply from an authorized reviewer carrying a one-time code; unauthorized attempts trigger a hard-stop alert.
Verified GAHAR references	Standard codes are cited only when retrieved from the official 2025 handbook; otherwise flagged unverified.
No reportability decision	For possible serious harm the system shows the 48-hour and 45-day GAHAR timelines; the quality officer decides what is reportable.
Fail safely	Unknown or conflicting classification goes to manual triage, never to low risk; failed runs are logged and alerted.

Known limits

- Not yet run end-to-end in a live n8n instance; no judge calibration or KPI actuals yet.
- Identifier detection is pattern-based and cannot recognize free-text personal names.
- No diagnosis, treatment advice, autonomous patient contact or definitive RCA without investigation.

What it takes to move toward a pilot

Step	Deliverable	Owner (proposed)
1. Controlled synthetic validation	Run 3 end-to-end scenarios, 10 timed reports, gold-labeled routing set; fill the KPI actuals.	Project team
2. Judge calibration	Two independent human raters score the same outputs; adjust rubric; document agreement.	Evaluation owner
3. Clinical governance review	Review prompts, hard-stop rules and approval roles with quality / clinical leadership.	Quality director
4. Security and privacy review	Persistent vector store, role-based access, retention policy, name-detection layer, mailbox hardening (SPF/DKIM).	IT / privacy officer
5. Approved pilot	One department, real workflow integration only after formal approval; confirm GAHAR edition and reporting channel.	Hospital management

Ask: approval to run the controlled synthetic validation and a governance review, as the gate to an approved pilot.

GAHAR basis and evidence statement

Item	Finding (researched 18 Oct 2026)
Standards edition	GAHAR Handbook for Hospital Accreditation Standards, 2025 edition (second edition; first was 2021); the national safety list is now called GAHAR Safety Requirements.
Serious / sentinel events	Notify GAHAR within 48 hours; submit the RCA and corrective plan within 45 days (handbook standard QPI.11; GAHAR procedure update of 12 April 2026).
Related manual	GAHAR Manual for Management of Reportable Events (2025).
Caution	Clause-level wording was read from search fragments; verify against the official PDFs before accreditation use.

Evidence statement. This pitch describes a proposal and an implementation blueprint. The n8n workflow was statically validated and its code unit-tested on sample inputs, but it has not been demonstrated in a live run. All metrics labeled target or assumption must be measured and replaced with actual evidence before final submission.